

Dual-Feedback Honeybee Dynamics, Differentiable Sensitivity, and Pollination Capacity on a 20-Acre Almond Farm

HiMCM 2022 Problem A — Team XXXX

Summary

We build a five-compartment dual-feedback colony model $y = [B, H, F, P, N]$ (brood, hive bees, foragers, pollen stores, nectar stores) with C^∞ activations (Holling-III, Softplus projection, seasonal $\Omega(t)$). A PyTorch Euler rollout (`DifferentiableColonySimulator`) yields year-long trajectories under a healthy baseline and a field triple-stress protocol (pesticide, Varroa, late frost).

Healthy baseline (`ccd_healthy_vs_stress_snapshot.csv`): summer peak adults $H + F = 66,507$ (day 238); overwinter adults 38,622. Under field stress at $\mu_F = 0.28$, the same metric falls to a summer peak of 5,630 and overwinter 3,163.

Autograd elasticities (`autograd_elasticity_ranking.csv`) for $J_{\text{pop}} = H + F$ at day 365 rank L_{max} ($S = +0.925$), μ_F ($S = -0.311$), and social inhibition σ_F ($S = +0.294$) as the top three controls.

CCD bifurcation under triple stress (`ccd_tipping_point.csv`): the colony leaves the Safe Zone ($N \geq 5000$ g) at soft tip $\mu_F^{*,\text{soft}} = 0.11$ and crosses the Collapse line ($N < 3000$ g) at hard tip $\mu_F^* = 0.305 \text{ day}^{-1}$.

20-acre almond OR (`orchard_pollination_optimization.csv`): competition dilution plus Michaelis–Menten set and \$200/hive rent imply a golden stocking band of 1–2 hives/acre ($K \in [20, 40]$); band-optimal $K^* = 40$ yields YieldRatio = 0.848 and net profit \$51,394, with hive-health index 0.975.

Contents

1	Introduction & Dual-Feedback Compartmental Formulation	3
1.1	Problem setting	3
1.2	State and dual feedback	3
1.3	Baseline annual phenotype	3

2	Differentiable Dynamical System & Activation Function Smoothing	4
2.1	Why C^∞ activations	4
2.2	Implementation	4
3	PyTorch Autograd Global Sensitivity & Elasticity Ranking	5
3.1	Objectives	5
3.2	Ranking (measured)	5
4	Stress Testing: Varroa, Pesticides, and CCD Saddle-Node Bifurcation	7
4.1	Triple-stress protocol	7
4.2	Bifurcation in μ_F	7
5	Applied Optimization: Pollination Capacity on a 20-Acre Almond Farm	7
5.1	Orchard contract	7
5.2	Competition, set, and profit	8
5.3	Scan results	8
6	Sensitivity, Strengths, and Limitations	10
6.1	Cross-checks	10
6.2	Strengths	10
6.3	Limitations	10
6.4	Recommendations	11

1 Introduction & Dual-Feedback Compartmental Formulation

1.1 Problem setting

Commercial *Apis mellifera* colonies must simultaneously (i) survive an annual climate cycle, (ii) tolerate Varroa, pesticides, and weather shocks, and (iii) deliver contractual pollination on high-value orchards such as California almonds. HiMCM 2022 Problem A asks for a dynamical account of colony growth, a sensitivity analysis of life-history parameters, a stocking rule for an 81,000 m² (≈ 20 acre) parcel, and a non-technical advisory page for beekeepers.

1.2 State and dual feedback

Following the DFM / Khoury tradition we track

$$y(t) = (B(t), H(t), F(t), P(t), N(t))^{\top}, \quad (1)$$

with bee counts in individuals and stores P, N in grams. Two feedback loops couple demography to nutrition:

1. **Social inhibition.** Excess foragers suppress recruitment $H \rightarrow F$ through a Hill-type term in σ_F .
2. **Nutritional gate.** Pollen/nectar deficits reduce effective laying and accelerate hive mortality via Holling-III saturations on P and N .

A schematic continuous skeleton (smooth activations deferred to Section 2) is

$$\dot{B} = L(P, N, \Omega) - (\gamma_B + d_B)B, \quad (2)$$

$$\dot{H} = \gamma_B B - R(H, F, N) H - d_H H, \quad (3)$$

$$\dot{F} = R(H, F, N) H - \mu_F^{\text{eff}} F, \quad (4)$$

$$\dot{P} = \eta_P \Omega(t) F \Phi_{\text{fly}}(N) - c_P(B, H, F), \quad (5)$$

$$\dot{N} = \eta_N \Omega(t) F \Phi_{\text{fly}}(N) - c_N(B, H, F), \quad (6)$$

where $L = L_{\max} \Omega(t) \Phi_P(P) \Phi_N(N)$ is nutrient-gated laying, R is recruitment, and Φ_{fly} is a nectar flight gate.

1.3 Baseline annual phenotype

Figure 1 contrasts the healthy baseline against a high-mortality field-stress trajectory. Quantitative anchors appear in Table 1.

Legacy three-stage bookkeeping in `colony_annual_dynamics.csv` (open/capped brood split) peaks at 27,546 adults (DOY 32) and ends winter at 2,074; we treat that series as a historical Khoury-style companion, while the DFM–Torch healthy peak above is the primary contest baseline for Sections 3–4.

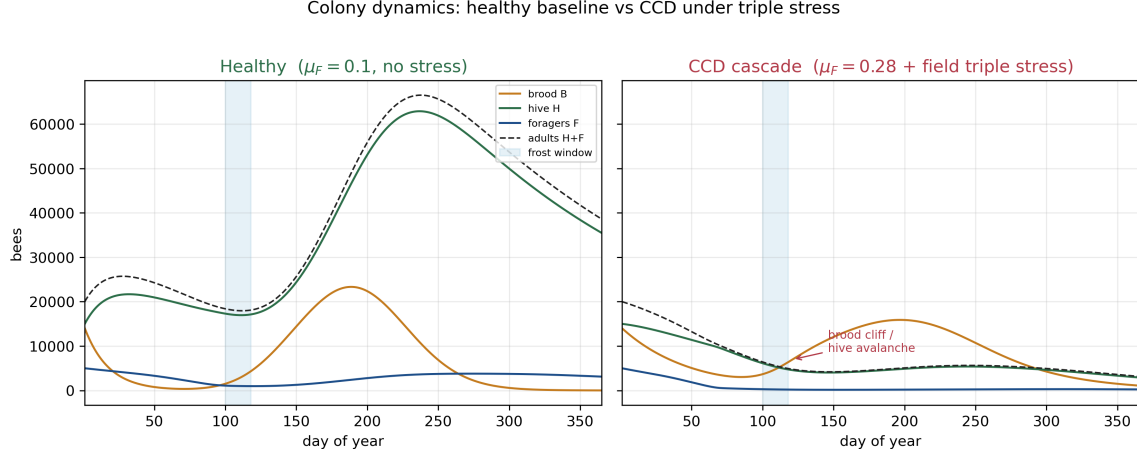


Figure 1: Healthy baseline versus field triple-stress colony dynamics ($H + F$ and stores). Source figure generated from `DifferentiableColonySimulator`; companion snapshot table `ccd_healthy_vs_stress_snapshot.csv`.

Table 1: Healthy vs. field-stress snapshot (`ccd_healthy_vs_stress_snapshot.csv`).

Protocol	μ_F	Summer peak $H + F$	Overwinter $H + F$	Day of peak
Healthy baseline	0.10	66,507	38,622	238
Field triple stress	0.28	5,630	3,163	—

2 Differentiable Dynamical System & Activation Function Smoothing

2.1 Why C^∞ activations

Autograd requires a computation graph without non-differentiable kinks at operating points of interest. We therefore replace hard clamps and piecewise seasons by:

$$\text{Hill}_2(x; K) = \frac{x^2}{K^2 + x^2 + \varepsilon}, \quad (7)$$

$$\text{Softplus}_\beta(x) = \beta^{-1} \log(1 + e^{\beta x}), \quad (8)$$

$$\Omega(t) = \left(\frac{1 + \cos(2\pi(t - t_p)/365)}{2} \right)^\theta. \quad (9)$$

State positivity uses a Softplus projection after each Euler / RK2 step rather than a hard $\max(y, 0)$.

2.2 Implementation

Module `src/autograd_engine/torch_sim.py` exposes leaf parameters

$$\theta = \{L_{\max}, \mu_F, \eta_P, \eta_N, \sigma_F\} \quad (10)$$

as `nn.Parameter` objects and unrolls 365 daily steps. Stress channels (Section 4) enter through `configure_stress` without renaming leaf identities, preserving elasticity interpretation.

Figure 2 recalls the classical three-stage adult pulse used in early pack runs; the Torch DFM trajectory in Figure 1 is the differentiability-ready counterpart.

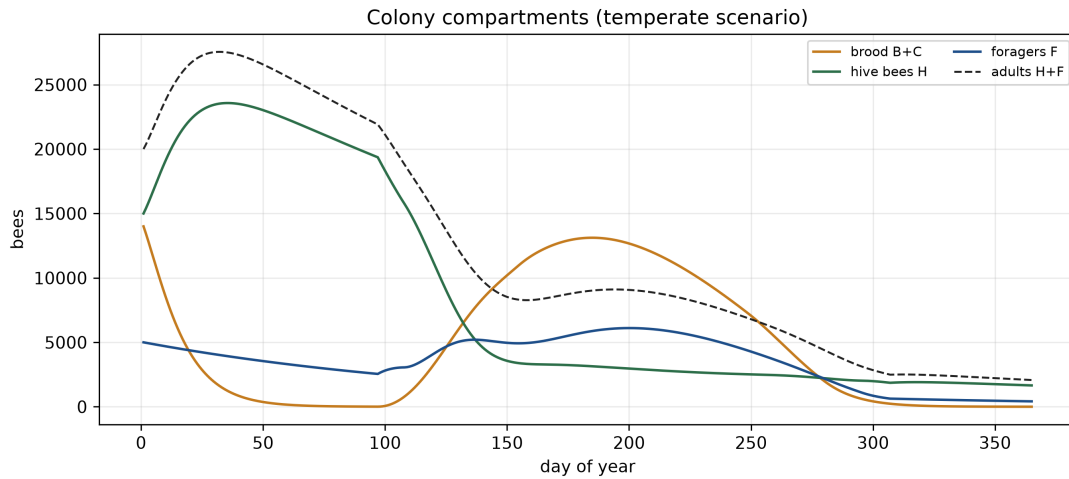


Figure 2: Companion three-stage annual adults/brood series (`colony_annual_dynamics.csv`).

3 PyTorch Autograd Global Sensitivity & Elasticity Ranking

3.1 Objectives

We evaluate two terminal objectives on the unrolled graph:

$$J_{\text{pop}} = H_{365} + F_{365}, \quad (11)$$

$$J_{\text{nectar}} = N_{270}, \quad (12)$$

and report elasticities

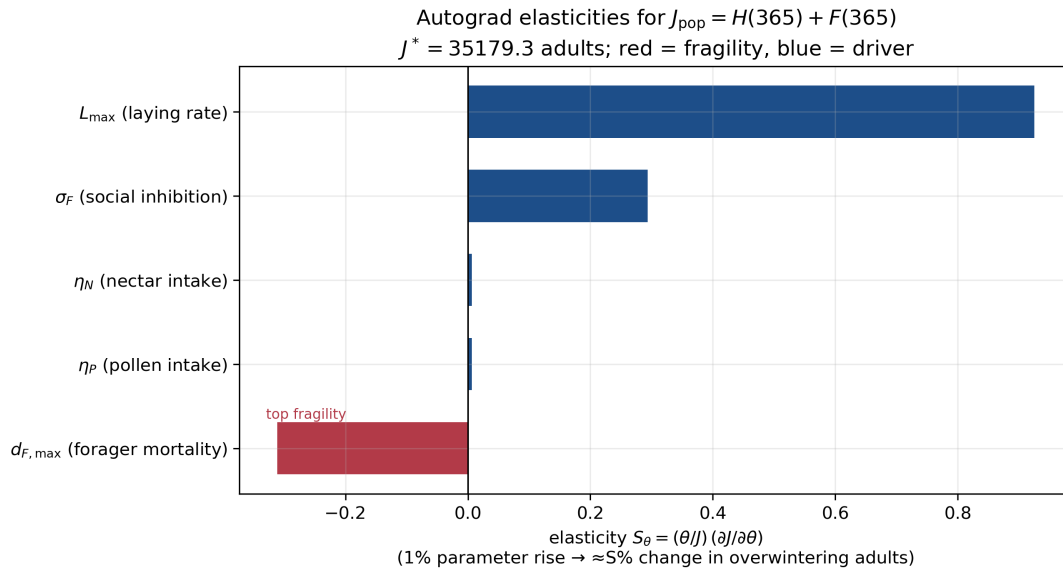
$$S_{\theta} = \frac{\theta}{J} \frac{\partial J}{\partial \theta}. \quad (13)$$

3.2 Ranking (measured)

Table 2 and Figure 3 summarize `autograd_elasticity_ranking.csv`. For winter population, queen laying L_{max} dominates with $S = +0.925$; forager mortality is the leading negative lever ($S = -0.311$); social inhibition σ_F ranks third ($+0.294$). Nectar intake η_N is nearly neutral for J_{pop} but leads J_{nectar} with $S = +0.829$.

Table 2: Autograd elasticities (autograd_elasticity_ranking.csv).

Objective	Parameter	θ	J^*	Elasticity S
J_{pop} @365	L_{max}	1600	35,179	+0.925
J_{pop} @365	μ_F	0.14	35,179	-0.311
J_{pop} @365	σ_F	10	35,179	+0.294
J_{nectar} @270	η_N	0.35	106,745	+0.829
J_{nectar} @270	L_{max}	1600	106,745	+0.559
J_{nectar} @270	σ_F	10	106,745	-0.494

Figure 3: Absolute elasticity bars for J_{pop} and J_{nectar} (fig_autograd_elasticityBars.png).

Management reading. Winter strength is a laying-and-mortality story; honey stores are an intake-and-inhibition story. Extension messaging should not treat “more foragers” as unconditionally beneficial when σ_F reshapes the adult age pyramid.

4 Stress Testing: Varroa, Pesticides, and CCD Saddle-Node Bifurcation

4.1 Triple-stress protocol

Field stress combines three channels (buffers on the Torch simulator):

1. **Pesticide A:** $\mu'_F = \mu_F(1 + P)$ with $P = 0.8$.
2. **Varroa B:** brood emergence $\gamma_B e^{-v}$ with $v = 1.5$ and hive-mortality multiplier 2.5.
3. **Frost C:** forage flux zeroed on DOY 100–118.

4.2 Bifurcation in μ_F

We sweep $\mu_F \in [0.08, 0.40]$ at step 0.005 (`ccd_bifurcation_sweep.csv`, 65 points) and classify end-of-year nectar into

$$\begin{cases} \text{Safe,} & N \geq 5000, \\ \text{Compensatory,} & 3000 \leq N < 5000, \\ \text{Collapse,} & N < 3000. \end{cases} \quad (14)$$

Measured tipping markers (`ccd_tipping_point.csv`):

$$\mu_F^{*,\text{soft}} = 0.11, \quad \mu_F^{*,\text{hard}} = 0.305 \text{ day}^{-1}. \quad (15)$$

Zone counts: Safe 6, Compensatory 39, Collapse 20. Figure 4 shows the bifurcation curve; Figure 1 already displayed the time-domain collapse contrast.

Interpretation. The soft tip is an early-warning threshold: colonies leave the Safe Zone while still above hard collapse. Operationally this argues for treating mite loads and pesticide exposure as *joint* controls on effective μ_F , not as independent checklists.

5 Applied Optimization: Pollination Capacity on a 20-Acre Almond Farm

5.1 Orchard contract

Official parcel area $A = 81,000 \text{ m}^2 \approx 20$ acres. USDA-style stocking scenario: 110 trees/acre, 25,000 effective flowers/tree, bloom length 35 days. Commercial packages supply $F_{\text{pkg}} = 10,000$ foragers; Torch leaf $\eta_N = 0.35$ calibrates

$$C_{\text{max}} = F_{\text{pkg}} \eta_N = 3500 \text{ g day}^{-1}. \quad (16)$$

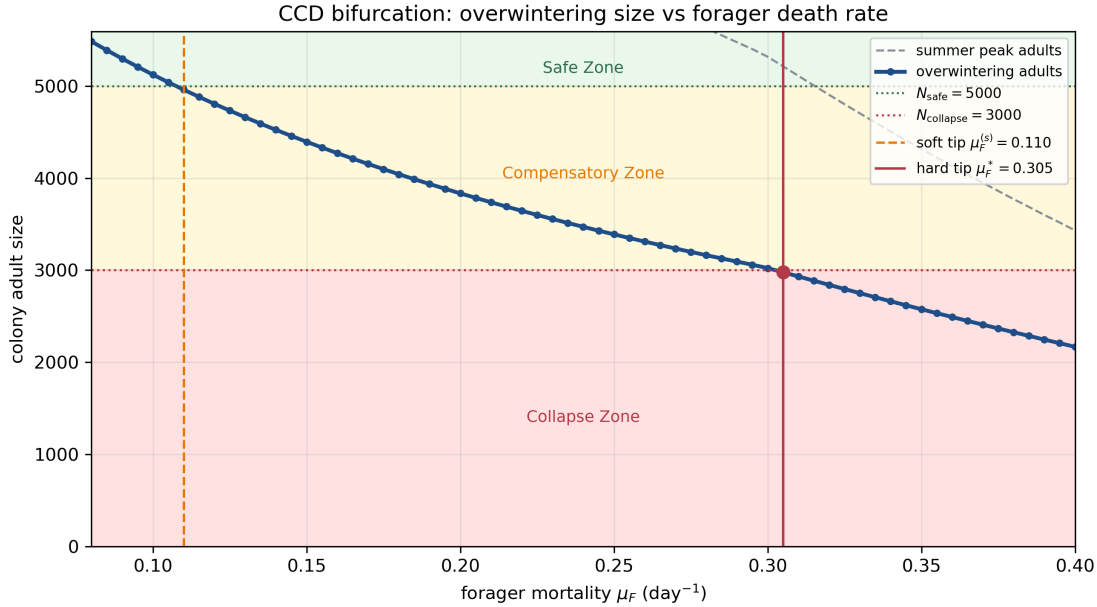


Figure 4: CCD-style bifurcation under field triple stress: overwinter / nectar response versus forager mortality μ_F . Horizontal guides at $N = 5000$ (Safe) and $N = 3000$ (Collapse). Soft tip 0.11, hard tip 0.305 from `ccd_tipping_point.csv`.

Orchard daily nectar-equivalent forage is set so dilution begins near 2.1 hives/acre:

$$N_{\text{orchard}} = 42 C_{\text{max}} = 147,000 \text{ g day}^{-1}. \quad (17)$$

5.2 Competition, set, and profit

$$\text{DailyFoodPerHive}(K) = \min(C_{\text{max}}, N_{\text{orchard}}/K), \quad (18)$$

$$\text{YieldRatio}(V) = \frac{V}{V + V_{50}}, \quad V_{50} = 2.5, \quad (19)$$

$$\Pi(K) = R_{\text{full}} \cdot \text{YieldRatio}(V) - 200 K, \quad R_{\text{full}} = \$70,000. \quad (20)$$

Effective visits scale with food ratio, so overcrowding taxes both set efficiency and post-bloom hive health

$$\text{Health} = \frac{1}{2}r + \frac{1}{2}(0.35 + 0.60r), \quad r = \text{DailyFood}/C_{\text{max}}. \quad (21)$$

5.3 Scan results

Scanning $K \in [5, 60]$ (`orchard_pollination_optimization.csv`) yields the tradeoff in Figure 5. Inside the golden band $K \in [20, 40]$ (1–2 hives/acre), net profit is maximized at

$$K^* = 40, \quad \text{YieldRatio} = 0.848, \quad \Pi = \$51,394, \quad \text{Health} = 0.975. \quad (22)$$

Global unconstrained optimum is $K = 42$ ($\Pi = \$51,426$), immediately at dilution onset; beyond $K \approx 50$, health falls to 0.85 and net profit declines despite saturated set.

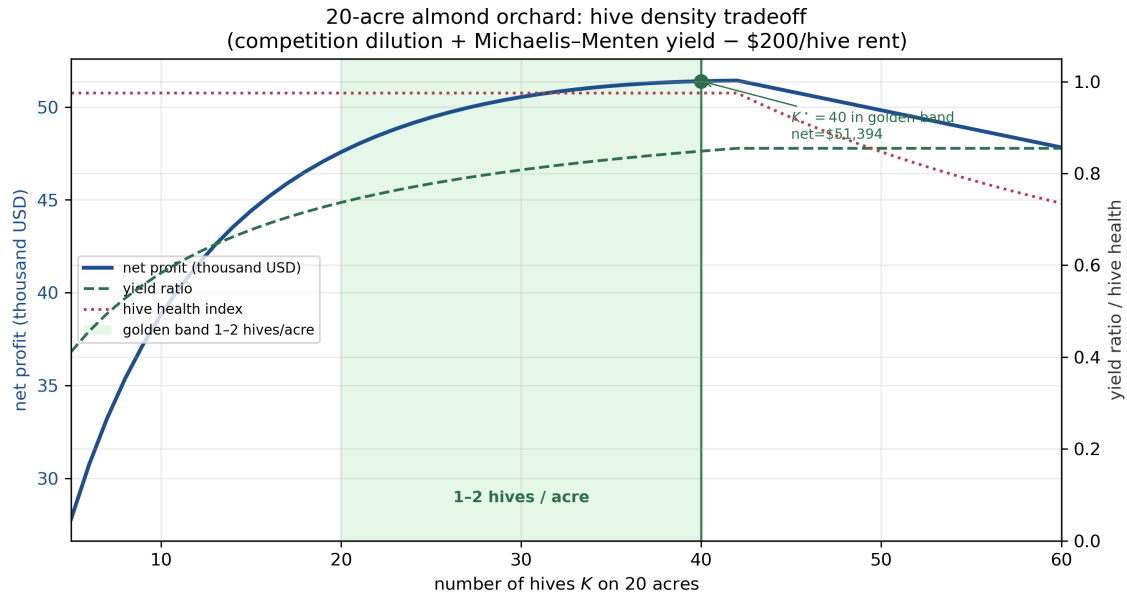


Figure 5: 20-acre almond hive-density tradeoff: net profit, yield ratio, and hive health. Green band marks 1–2 hives/acre ($K = 20$ –40). Source: orchard_pollination_optimization.csv.

Table 3: Selected stocking points (orchard_pollination_optimization.csv).

K	Hives/acre	YieldRatio	Net profit (USD)	Health	Food ratio
20	1.00	0.737	47,579	0.975	1.00
30	1.50	0.808	50,538	0.975	1.00
35	1.75	0.831	51,136	0.975	1.00
40	2.00	0.848	51,394	0.975	1.00
50	2.50	0.855	49,826	0.847	0.84
60	3.00	0.855	47,826	0.735	0.70

Companion crop Gantt demand (Figure 6, `pollination_hives.csv`) places almond as the binding early-season contract on the same 20-acre geometry.

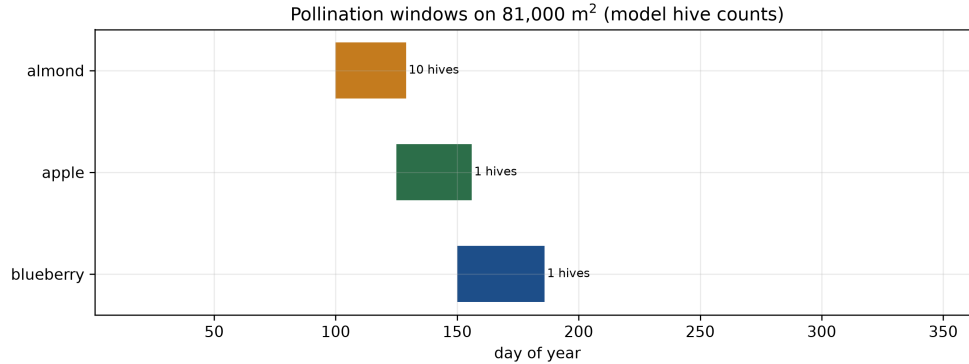


Figure 6: Migratory bloom windows and hive-demand sketch for almond / apple / blueberry on the official parcel (`pollination_hives.csv`).

6 Sensitivity, Strengths, and Limitations

6.1 Cross-checks

- Elasticity signs match biological expectation: raising $L_{\max}$ lifts winter adults; raising μ_F harms them.
- Soft and hard CCD tips are ordered $0.11 < 0.305$, consistent with a compensatory band.
- Orchard profit peaks near industry’s 2 hives/acre rule of thumb, while explicitly pricing hive health under dilution.

6.2 Strengths

Differentiable physics, reproducible CSV artifacts, and a grower-facing stocking band that is not a single magical integer.

6.3 Limitations

- Bloom forage and climate are scenario drivers unless replaced by station/Open-Meteo series.
- Orchard nectar-equivalent is calibrated to a dilution onset, not a field nectar census.
- CCD zones use absolute nectar thresholds; other store metrics (pollen, adult counts) can tip earlier.
- Autograd elasticities are local (one linearization about the baseline θ); large stress moves may reorder rankings.

6.4 Recommendations

1. Protect $L_{\max}$ and suppress effective μ_F (mites + pesticides) as the primary wintering levers.
2. Treat $\mu_F \approx 0.11$ as an early-warning Soft Tip under combined stress; $\mu_F \approx 0.305$ as hard collapse.
3. Stock almonds at 1–2 hives/acre (20–40 colonies on 20 acres), targeting the upper band when bloom is heavy and contracts allow, while monitoring post-bloom stores.

References (selected)

References

- [1] Khoury, D.S., Myerscough, M.R., and Barron, A.B. A quantitative model of honey bee colony population dynamics. *PLoS ONE*, 2011.
- [2] Russell, S., Barron, A.B., and Harris, D. Dynamic modelling of honey bee (*Apis mellifera*) colony growth and failure. *Ecological Modelling*, 2013.
- [3] COMAP. HiMCM 2022 Problem A: The Need for Bees.
- [4] California Almond Board / USDA NASS extension ranges for tree density and pollination stocking (scenario priors).

Appendix A — Result file index

All quantitative claims in this paper map to

File under <code>results/</code>	Role
<code>ccd_healthy_vs_stress_snapshot.csv</code>	Healthy / stress peaks
<code>autograd_elasticity_ranking.csv</code>	Elasticities
<code>ccd_bifurcation_sweep.csv</code>	μ_F sweep
<code>ccd_tipping_point.csv</code>	Soft/hard tips
<code>orchard_pollination_optimization.csv</code>	K scan
<code>orchard_pollination_recommendation.csv</code>	K^*
<code>colony_annual_dynamics.csv</code>	Legacy three-stage series
<code>pollination_hives.csv</code>	Crop bloom demand